Software Tech 2
7170
Tutorial 4
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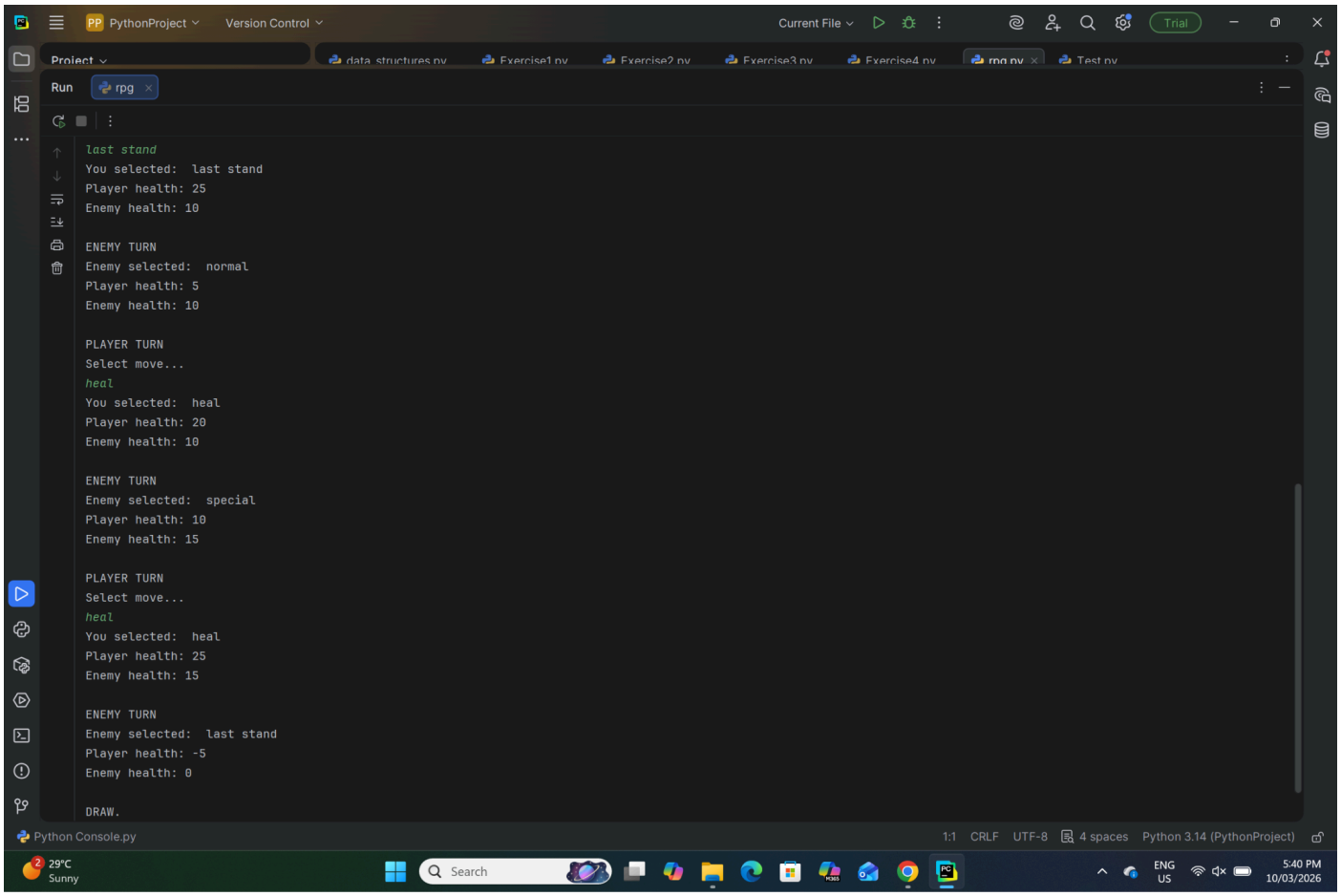
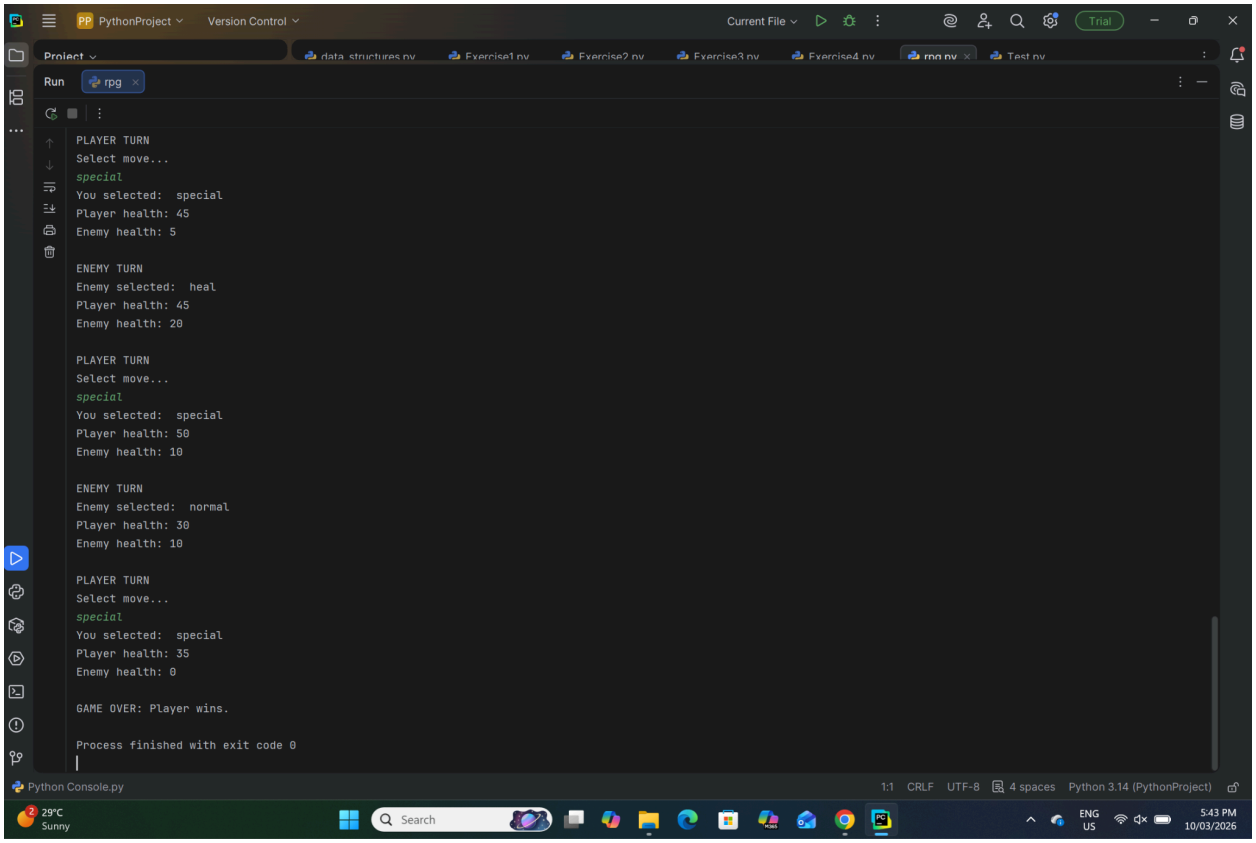
Task 1:

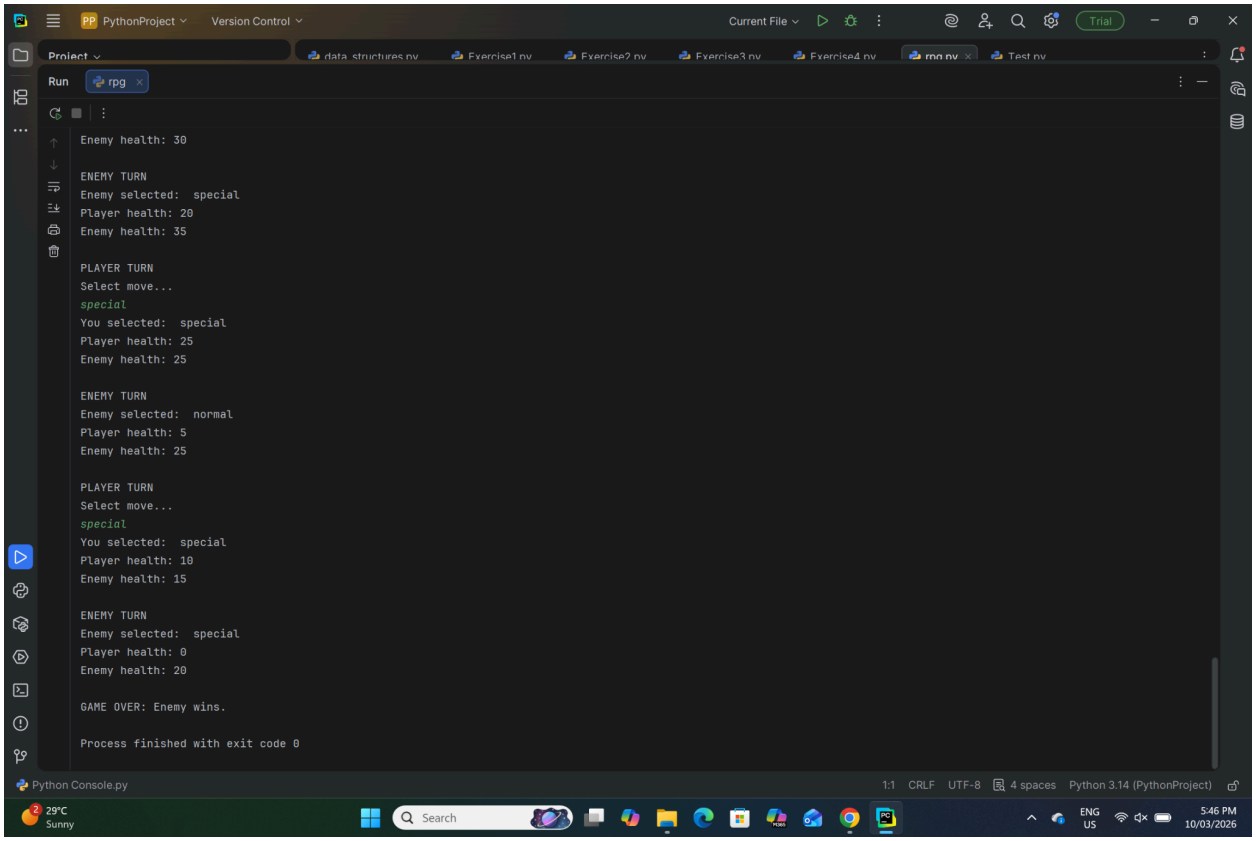
Test output no.1: –Draw–



Test output no.2: –Player Wins–



Test output #3: –Enemy Wins–



Task 3

1) Linear Search Algorithm and show the algorithm efficiency for different data sizes

With a very large number linear search time is efficient but the higher the number in the list the longer that it will take

Linear search execution time: 0.339076 seconds
99999999999

- The algorithm will check everything until it reaches that number

2)

Binary search is a lot faster rather than linear due to the fact of it not searching every input like linear binary skips over numbers, however binary only works on a sorted list

3)

Binary is faster than linear search on a sorted list because it repeatedly halves the inputs and searches it while linear searches every input until it finds its location

4) Test Example Of changing Values in Insertion

```
Linear search execution time: 1.273686 seconds
Binary search execution time: 0.000705 seconds
Sorted array: [11, 12, 22, 25, 64]
Sorted array: [11, 22, 34, 80, 112]
Bubble Sort Execution Time: 10.655388 seconds
Insertion Sort Execution Time: 4.267906 seconds
Binary Search (With Sorting) Execution Time: 5.469620 seconds
Linear Search Execution Time: 0.000210 seconds
```

5) *Bubble Sort Test*

```
C:\Users\Ryano\PycharmProjects\PythonProject\.venv\Sc
99999
-1
1
8
Linear search execution time: 0.524689 seconds
Binary search execution time: 0.000680 seconds
Sorted array: [10, 30, 100, 111, 400]
Sorted array: [1, 11, 12, 22, 64]
Bubble Sort Execution Time: 10.296867 seconds
Insertion Sort Execution Time: 4.070942 seconds
```

6) *Runtimes*

Bubble Sort 10.2s
Insertion 4.2s

Bubble sort does swaps over time which makes it slower. Insertion sort places it into the correct position

Task 4

```
Data size: 100
Insertion Sort + Linear Search: 0.000494 seconds
Bubble Sort + Linear Search: 0.001003 seconds
Insertion Sort + Binary Search: 0.000482 seconds
Bubble Sort + Binary Search: 0.000937 seconds

Data size: 1000
Insertion Sort + Linear Search: 0.038723 seconds
Bubble Sort + Linear Search: 0.085675 seconds
Insertion Sort + Binary Search: 0.041531 seconds
Bubble Sort + Binary Search: 0.104062 seconds

Data size: 5000
Insertion Sort + Linear Search: 1.060360 seconds
Bubble Sort + Linear Search: 2.479606 seconds
Insertion Sort + Binary Search: 1.075511 seconds
Bubble Sort + Binary Search: 2.427628 seconds
```